

AI in Medicine

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Disease Diagnosis and Prediction

The integration of AI in disease diagnosis and prediction is focused on enhancing the accuracy and speed of identifying medical conditions through various predictive models. Key research directions include:

Predictive Modeling: Developing AI models that leverage electronic health records (EHRs) and other data sources to predict disease onset and progression. Recent studies have shown that AI can identify risk factors for diseases like diabetes and cardiovascular conditions by analyzing a broad set of patient data.

Machine Learning Techniques: Utilizing advanced machine learning techniques, such as deep learning and ensemble methods, to improve diagnostic accuracy. Research indicates that deep learning models can outperform traditional algorithms in predicting outcomes based on complex datasets.

Early Identification: Focusing on algorithms designed to detect early signs of diseases, which can lead to timely interventions. This approach is particularly crucial in conditions such as cancer, where early detection significantly improves treatment outcomes.



Medical Imaging Analysis

AI's impact is particularly pronounced in medical imaging, where it enhances diagnostic accuracy and workflow efficiency. Major research directions in this area encompass:

Deep Learning Applications: The use of convolutional neural networks (CNNs) and other deep learning models for image classification and segmentation. Studies demonstrate that AI can analyze medical images (e.g., X-rays, MRIs) with accuracy comparable to or better than human specialists, thereby assisting in disease detection and treatment planning.

Image Segmentation: Advanced techniques for accurately segmenting anatomical structures and pathological features in medical images. Effective segmentation is vital for aligning treatment plans with specific anatomical details and improving surgical outcomes.

Multimodal Imaging Integration: Combining data from different imaging modalities to create a more comprehensive view of a patient's condition. Research suggests that integrating datasets from CT, MRI, and PET scans can enhance diagnostic precision and aid in complex disease assessments.



Research and Data Analysis

The transformative power of AI in healthcare heavily relies on robust data analysis capabilities. This field's primary research directions include:

EHR Data utilization: Leveraging large datasets from electronic health records to develop insights into patient populations, treatment efficacy, and disease trends. AI models are being refined to extract actionable insights that can enhance clinical decision-making and patient management.

Population Health Analytics: Researching how AI can analyze population health data to identify trends, predict outbreaks, and improve public health responses. The focus is on using diverse datasets to inform healthcare policies and optimize resource allocation.

Integration of Genomic Data: Combining genomic data with clinical data to better understand disease mechanisms and personalize treatment strategies. The integration of genetic and phenotypic information enhances predictive capabilities and supports the development of precision medicine approaches.



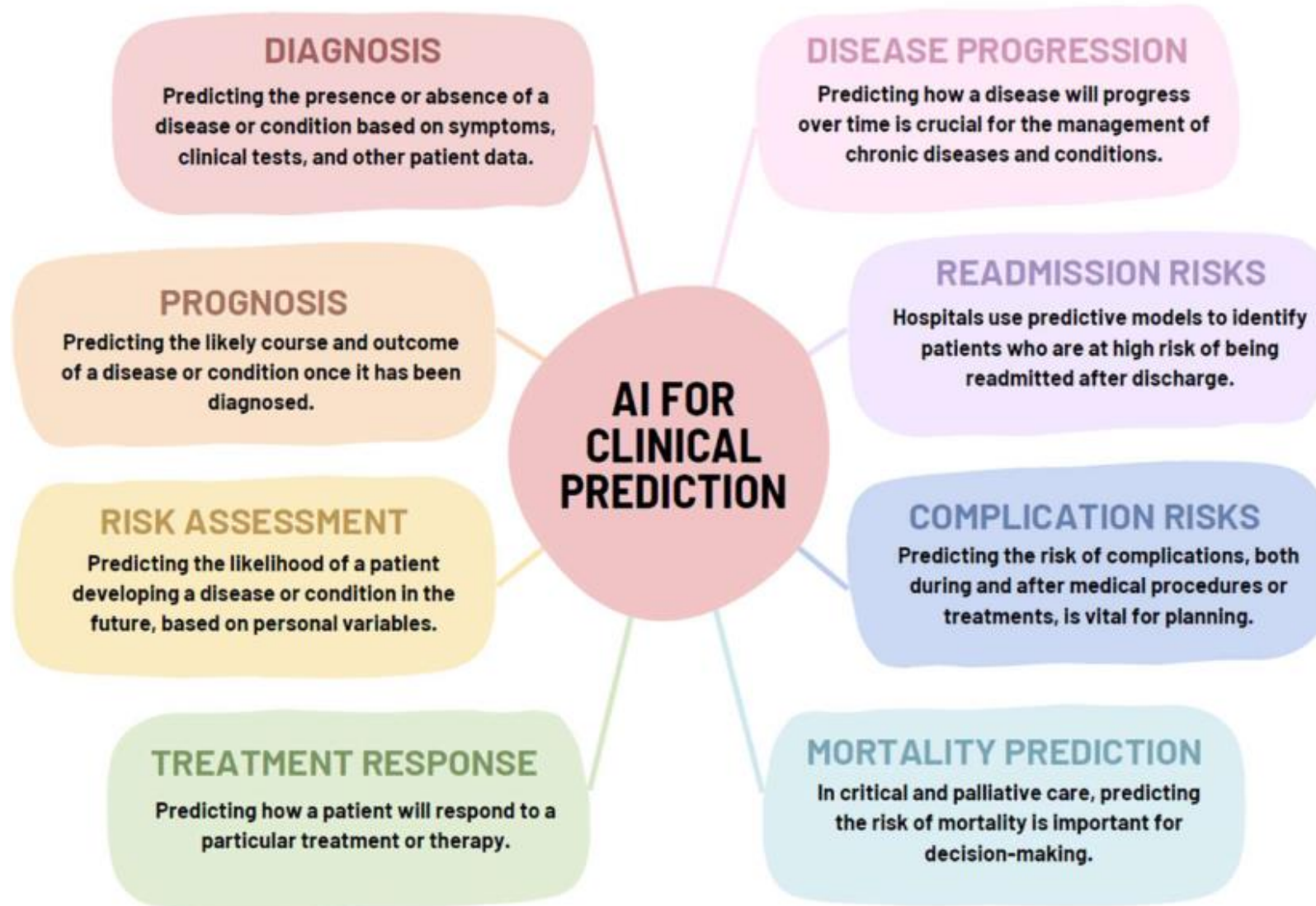
DISEASE DIAGNOSIS AND PREDICTION



ARTIFICIAL INTELLIGENCE FOR CLINICAL PREDICTION: EXPLORING KEY DOMAINS AND ESSENTIAL FUNCTIONS

- <https://doi.org/10.1016/j.cmpbup.2024.100148>

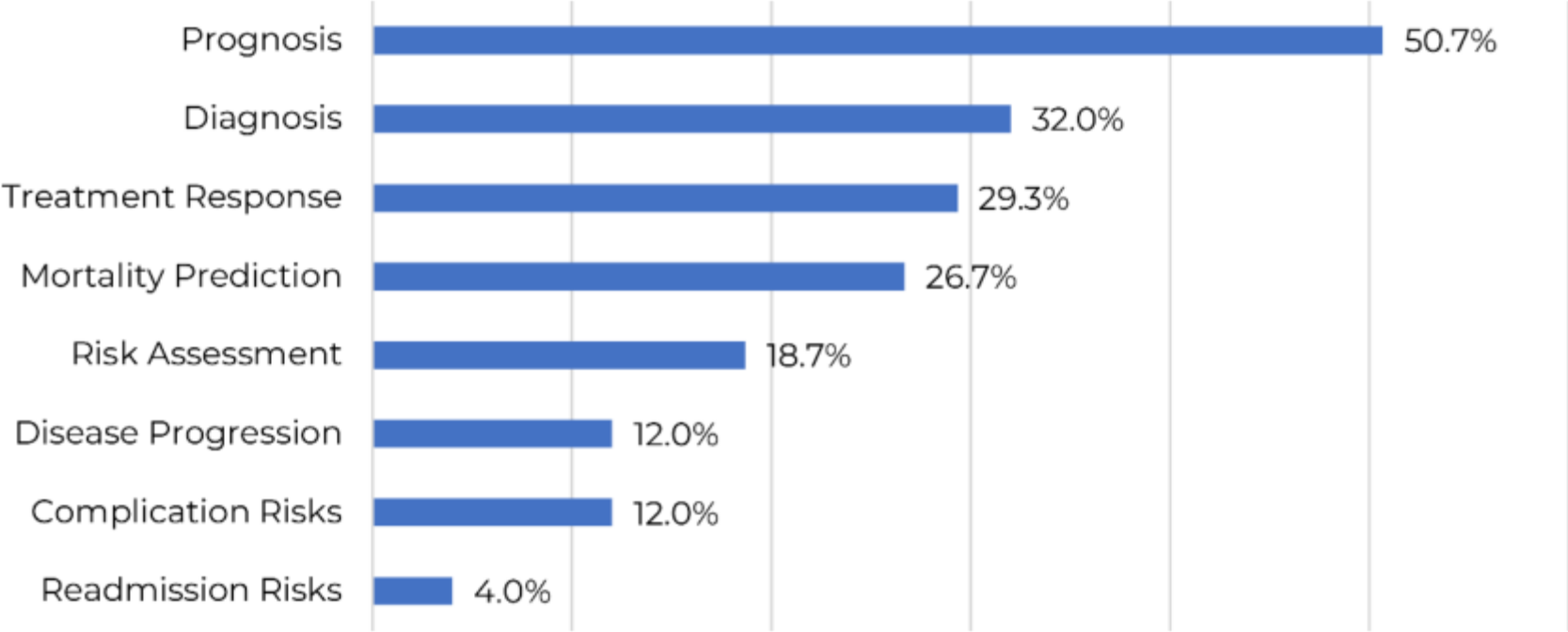
AI SUPPORTS EIGHT DOMAINS OF CLINICAL PREDICTION



THE AI EIGHT DOMAINS FOR CLINICAL PREDICTION

SN	Domains	Functions			
1	Diagnosis	Predicting the presence or absence of a disease or condition based on symptoms, clinical tests, and other patient data. Diagnostic predictions are essential for early detection and timely treatment.	5	Disease Progression	Predicting how a disease will progress over time is crucial for chronic diseases such as diabetes, heart disease, and neurological disorders. It helps in planning long-term treatment strategies and anticipating future care needs.
2	Prognosis	Predicting the likely course and outcome of a disease or condition once it has been diagnosed. Prognostic predictions help in understanding the disease's progression, potential complications, and the likely response to treatment.	6	Readmission Risks	Hospitals and healthcare providers use predictive models to identify patients who are at high risk of being readmitted after discharge. This helps in providing targeted interventions to reduce readmission rates.
3	Risk Assessment	Predicting the likelihood of a patient developing a disease or condition in the future based on various risk factors such as genetics, lifestyle, environmental exposures, and existing health conditions.	7	Complication Risks	Predicting the risk of complications, both during and after medical procedures or treatments, is vital for informed consent and for planning to mitigate those risks.
4	Treatment Response	Predicting how a patient will respond to a particular treatment or therapy. This is especially important in personalised medicine, where the goal is to tailor treatments to individual patients for maximum effectiveness.	8	Mortality Prediction	In critical care and palliative care settings, predicting the risk of mortality is important for decision-making regarding treatment intensity and end-of-life care planning.

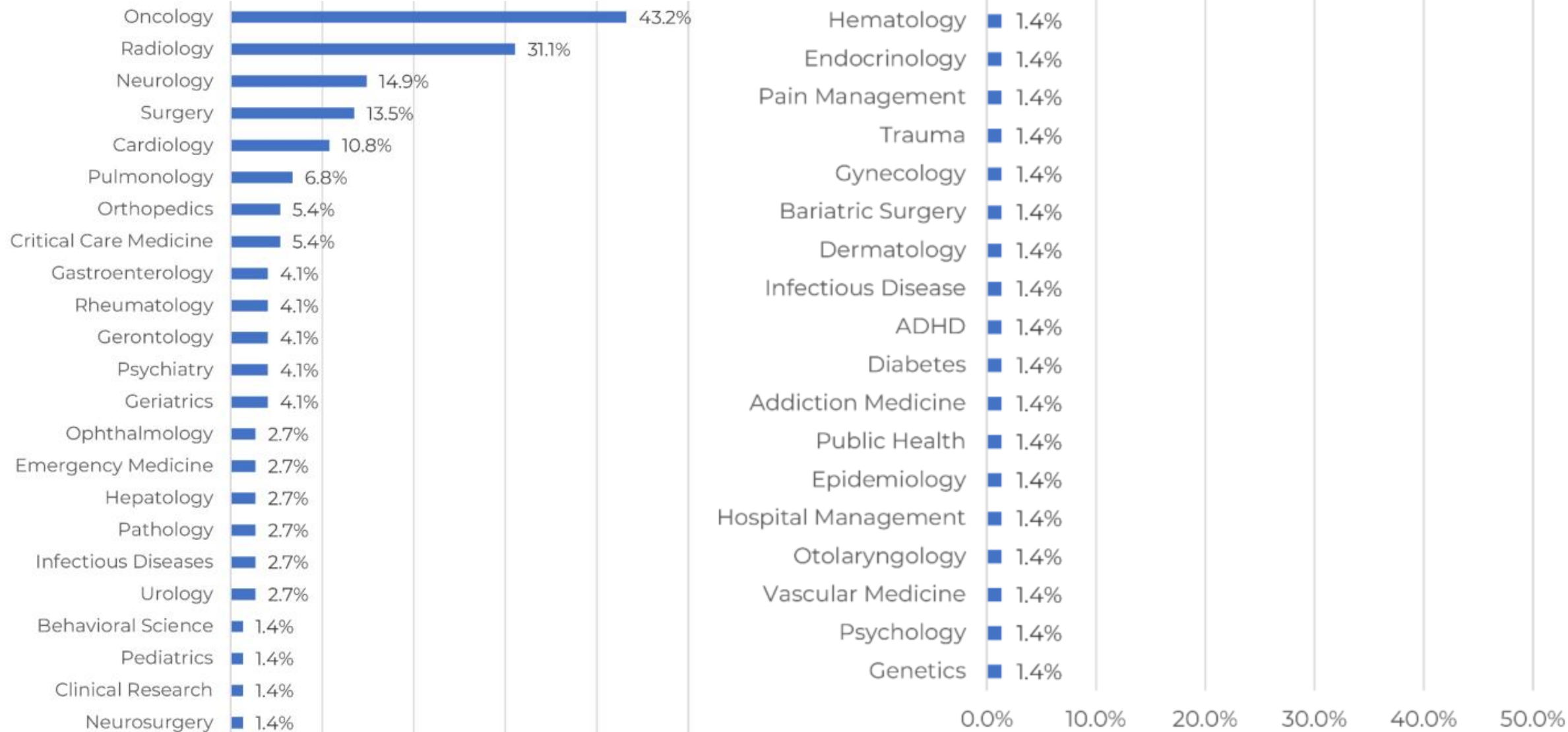
AI CONTRIBUTION TO CLINICAL PREDICTION DOMAINS, BASED ON THE 74 STUDIES



SPECIALTIES WHERE AI SUPPORTS CLINICAL PREDICTION, BASED ON THE 74 STUDIES

Speciality	Number of studies		
Oncology	32	Psychology	1
Radiology	23	Vascular medicine	1
Neurology	11	Otolaryngology	1
Surgery	10	Hospital management	1
Cardiology	8	Epidemiology	1
Pulmonology	5	Public health	1
Critical care medicine	4	Addiction medicine	1
Orthopaedics	4	Diabetes	1
Geriatrics	3	ADHD	1
Psychiatry	3	Infectious disease	1
Gerontology	3	Dermatology	1
Rheumatology	3	Bariatric surgery	1
Gastroenterology	3	Gynaecology	1
Urology	2	Trauma	1
Infectious diseases	2	Pain management	1
Pathology	2	Endocrinology	1
Hepatology	2	Haematology	1
Emergency medicine	2	Neurosurgery	1
Ophthalmology	2	Clinical research	1
Genetics	1	Paediatrics	1
		Behavioural science	1

SPECIALTIES WHERE AI SUPPORTS CLINICAL PREDICTION, BASED ON THE 74 STUDIES



RECOMMENDATIONS

Based on the findings of this review, the following eight recommendations are proposed to optimize the integration of AI in clinical prediction:

- 1. Enhance Data Quality and Accessibility:** Ensure the collection of high-quality, diverse, and comprehensive healthcare data to train AI models effectively. Improved data sharing protocols should be established to facilitate access to diverse datasets while maintaining patient privacy and data security.
- 2. Promote Interdisciplinary Collaboration:** Encourage collaboration between AI experts, healthcare professionals, and researchers. This collaboration will foster the development of AI tools that are clinically relevant and user-friendly for healthcare providers.
- 3. Focus on Ethical and Transparent AI Practices:** Develop AI systems with ethical considerations and transparency. This includes addressing biases in AI algorithms, ensuring equitable healthcare delivery, and maintaining transparency in AI decision-making processes.
- 4. Invest in AI Education and Training:** Provide education and training for healthcare professionals on AI tools and their applications in clinical prediction. This will enhance their ability to interpret AI-generated predictions and integrate them into clinical practice.

RECOMMENDATIONS

5. **Expand Clinical Trials and Research:** Conduct extensive clinical trials to validate the efficacy and safety of AI applications in clinical prediction. Further research should also focus on exploring the potential of AI in unexplored areas of healthcare.
6. **Regulatory Oversight and Guidelines:** Develop and implement regulatory frameworks and guidelines to govern the use of AI in clinical prediction. This will ensure that AI tools meet quality and safety standards before their clinical implementation.
7. **Patient Engagement and Consent:** Involve patients in the AI integration process, ensuring they are informed about how AI is used in their care. Patient consent should be sought, especially in cases where AI influences critical healthcare decisions.
8. **Continuous Monitoring and Improvement:** Regularly monitor the performance of AI systems in clinical settings and make iterative improvements based on feedback from healthcare providers and patients.

ARTIFICIAL INTELLIGENCE IN DISEASE PREDICTION USING REAL WORLD HEALTH DATA: TRENDS, CHALLENGES, AND THE FUTURE

Recent years have seen significant progress in using artificial intelligence (AI) to develop disease prediction models, which have the potential to improve diagnosis precision, enable early disease prevention, streamline clinical decision making, and reduce healthcare costs. This progress has been supported by the availability of large and diverse biomedical data, including Electronic Health Records (EHRs), which have become a valuable resource for disease prediction. The wide adoption of structured EHR systems has enabled historical patient records to serve as one of the most valuable resources for disease prediction. Traditional research for building disease prediction models relied on experts' ability to define the appropriate features and design the model's structure. With massive developments in the field of AI, the applications of deep learning on structured EHR escalated – the main catalysts were its ability to process large amounts of data and learn complex features. The growing advancements in AI and DL has significantly evolved the thought process of the researchers when building predictive models for disease. In this presentation, I will talk about how the process of overcoming the challenges of EHR data has led to adoption of various advanced DL models for disease prediction. I will share my experiences on the overall experiments we conducted in our efforts to build advanced DL models for predicting diabetes complications and outline future research directions.

POTENTIAL FOR A PREDICTIVE MODEL TO AID IN THE EARLY IDENTIFICATION OF LIVER DISEASE

Liver disease is potentially curable if identified early and treated appropriately. However, this disease often goes unnoticed until a liver transplant is the only option. **An AI-based predictive model to help identify patients at risk of severe liver disease** could play a crucial role in early diagnosis of liver cancer. In association with other clinically relevant information, a predictive model could potentially enable early intervention and help to avoid progression to cirrhosis, liver failure, the need for liver transplant, and even mortality.



MEDICAL IMAGING ANALYSIS



AI IN MEDICAL IMAGE ANALYSIS: REDEFINING DIAGNOSTIC ACCURACY AND SPEED

Medical imaging technologies such as X-rays, MRIs, and CT scans have been crucial in diagnosis for years. AI introduces a paradigm shift by improving diagnostic accuracy, streamlining analysis processes, and reducing errors made by humans. The potential of AI not only benefits patients by enhancing outcomes but also optimizes operational workflows making it an essential tool for forward-thinking healthcare organizations.

[AI in Medical Image Analysis: Redefining Diagnostic Accuracy and Speed](#)

AI IN MEDICAL IMAGE ANALYSIS: REDEFINING DIAGNOSTIC ACCURACY AND SPEED

Role of AI in Various Types of Medical Imaging Modalities

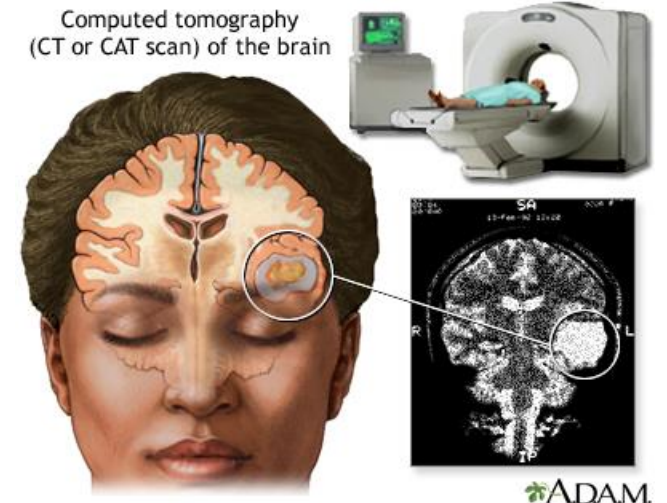
1. X-Rays:

- X-rays or radiographs are one of the basic tools of medical imaging. The images obtained are very clear, particularly in an X-ray of dense structures such as the bones, so it is easy and quick to identify a broken bone, joint dislocation, or infection.



2. Computed Tomography (CT) Scans:

- CT scans are important in modern diagnosis, especially in detecting cancer, heart conditions, internal injuries, and bleeding. CT scans provide detailed cross-sectional images of the body. This level of detail helps with the accurate staging, assessment of the heart and evaluation of traumatic injuries.

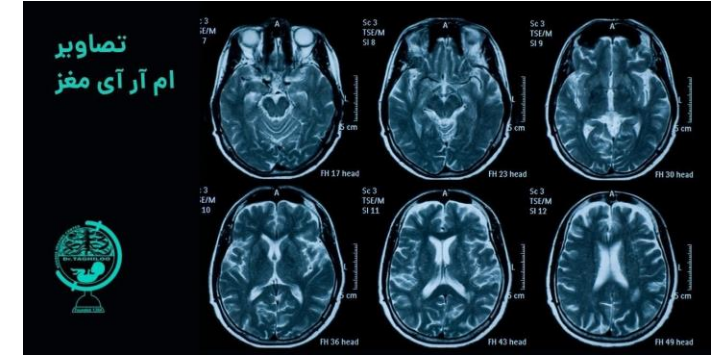


AI IN MEDICAL IMAGE ANALYSIS: REDEFINING DIAGNOSTIC ACCURACY AND SPEED

Role of AI in Various Types of Medical Imaging Modalities

3. Magnetic Resonance Imaging (MRI):

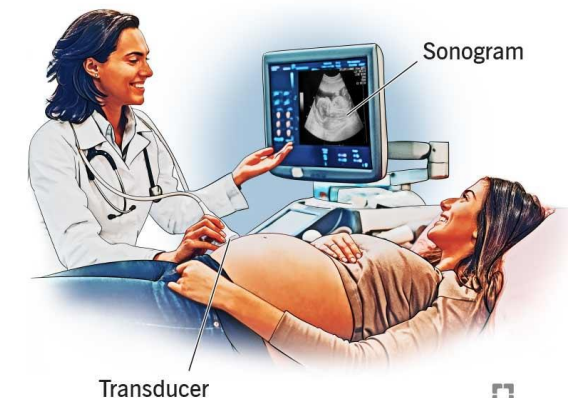
- MRI is the flexible and effective large-scale diagnostic tool commonly applied in brain imaging, checking the injury to the spinal cord, and diagnosing the diseases in joints. Compared to other imaging techniques, MRI captures extremely precise images of organs, soft tissues, and bones without ionizing radiation. Thus, it represents the most beneficial application to illustrate complex structures, for instance, the brain, muscles, and ligaments, with unmatched clarity and detail.



4. Ultrasound:

- CT scans are important in modern diagnosis, especially in detecting cancer, heart conditions, internal injuries, and bleeding. CT scans provide detailed cross-sectional images of the body. This level of detail helps with the accurate staging, assessment of the heart and evaluation of traumatic injuries.

Abdominal ultrasound

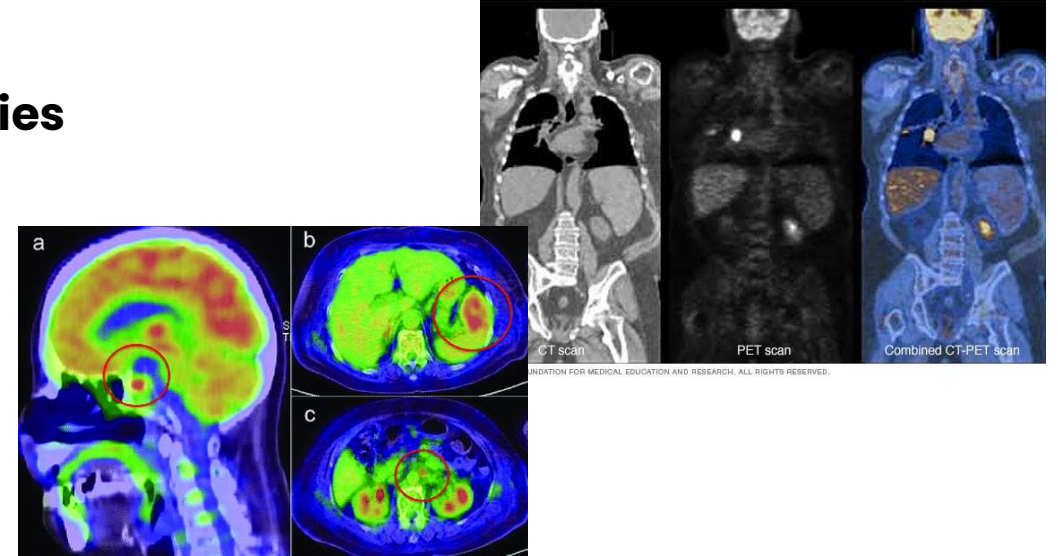


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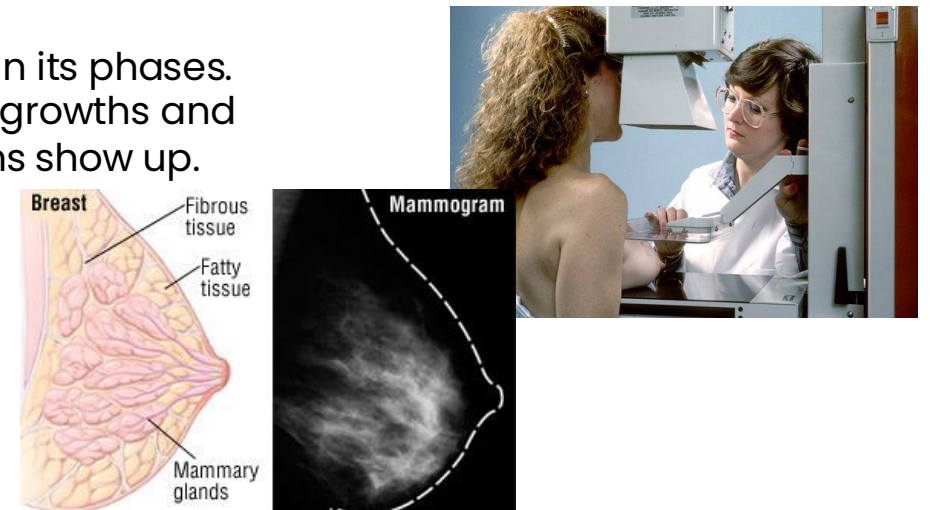
5. Positron Emission Tomography (PET) Scans:

- PET scans are very useful for imaging the metabolic activity occurring in tissues and organs making them a crucial tool in fields such as oncology, neuroscience, and cardiology.



6. Mammography:

- Mammography is an imaging method used to spot breast cancer in its phases. Mammograms, which provide images of breast tissue can identify growths and alterations that could suggest cancer well before any physical signs show up.

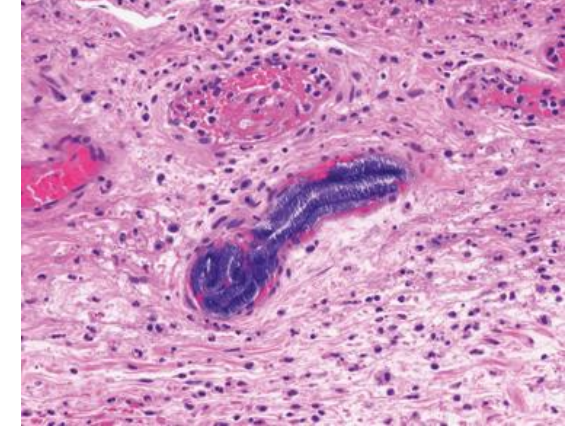


AI IN MEDICAL IMAGE ANALYSIS: REDEFINING DIAGNOSTIC ACCURACY AND SPEED

Role of AI in Various Types of Medical Imaging Modalities

7. Pathology Slides:

- Pathology slides are used to diagnose cancer and other cellular diseases. They offer high-resolution digital images for thorough tissue investigation. This microscopical assessment is critical for detecting cellular abnormalities and disease indicators.



8. Dermatological Images:

- Dermatological images are critical for identifying skin cancer and other dermatological conditions. These images are gathered in non-invasive ways, such as photographs of skin lesions, rashes, and other disorders, making the procedure comfortable and safe for patients. Dermatologists can use these images to visually diagnose numerous skin diseases without having to do invasive procedures.



AI IN MEDICAL IMAGE ANALYSIS: REDEFINING DIAGNOSTIC ACCURACY AND SPEED

Role of AI in Various Types of Medical Imaging Modalities

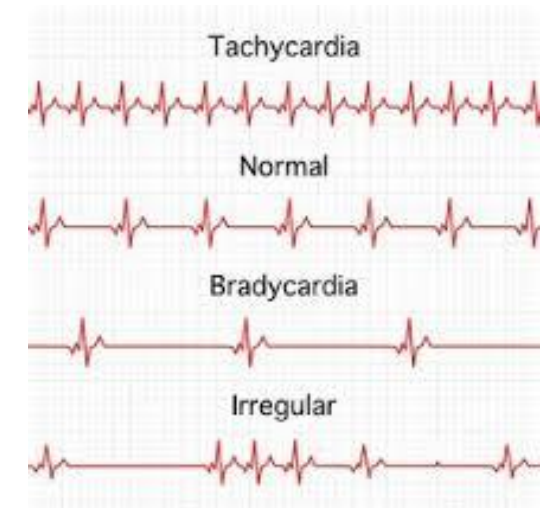
9. Ophthalmological Images:

- Ophthalmological images are essential for identifying a variety of eye diseases, including diabetic retinopathy, glaucoma, and macular degeneration. These images provide precise views of the retina and other eye components, allowing healthcare professionals to pinpoint abnormalities and disease signs with high precision. These provide the clarity and detail required for accurate diagnosis and effective monitoring of eye health.



10. Electrocardiogram (ECG) traces:

- ECG traces are critical for detecting heart abnormalities and conditions, as they provide a quick and non-invasive way to monitor the heart's electrical activity. ECG readings offer important information about heart rhythms, aiding in detecting arrhythmias, heart attacks, and other cardiac conditions. Due to their ease of use and speed, ECG tests have become a practice in emergencies and regular cardiac care.

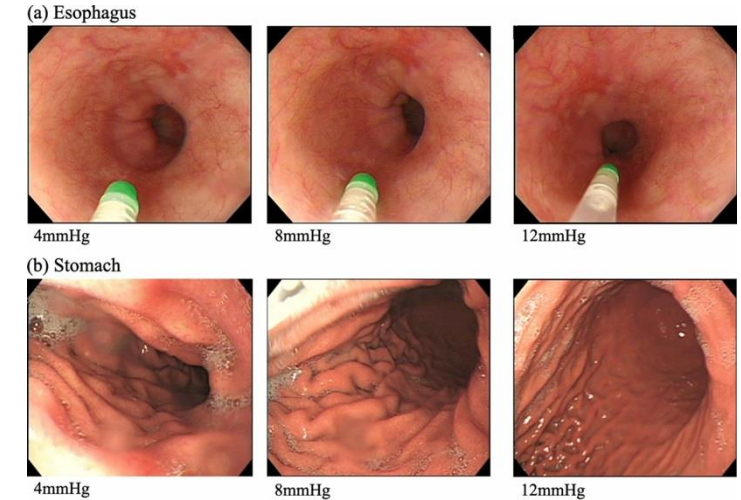


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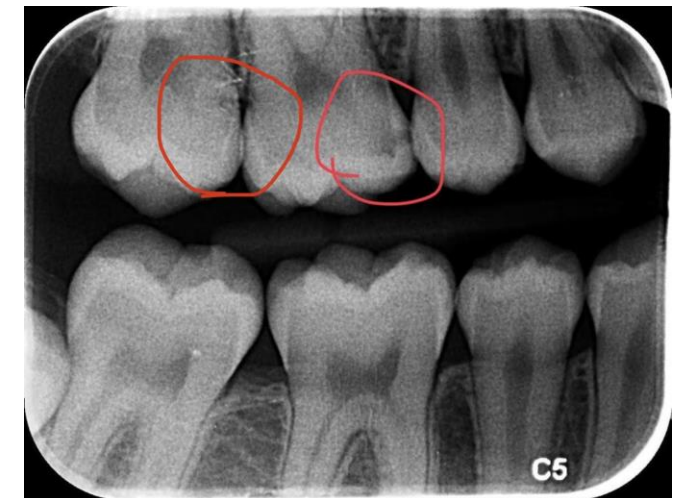
9. Endoscopic images:

- Endoscopic images are imperative for examining the internal surfaces of organs such as the gastrointestinal tract and airways. The images provide a direct vision of the inner structures, thus allowing healthcare workers to detect anomalies such as ulcers, polyps, tumors, and blockages in real time.



10. Dental X-rays:

- Dental X-rays play a key role in health assessments as they uncover cavities, bone issues, and other oral conditions that may go unnoticed during routine checkups. These X-ray images provide views of teeth, bones, and surrounding tissues, aiding dentists in problem detection and treatment planning.



IMAGING ANALYSIS, TECHNIQUES, & AI

Research in radiological sciences, particularly in the areas of imaging analysis, techniques, and artificial intelligence (AI), is dynamic and continuously evolving. Numerous studies and developments are contributing to advancements in the field. Here are some key areas of research:

1. Deep Learning and Convolutional Neural Networks (CNNs):

- Researchers are exploring the application of deep learning techniques, especially CNNs, for image classification, segmentation, and detection in various modalities such as CT, MRI, and mammography.
- Transfer learning, where pre-trained models are fine-tuned for specific radiological tasks, is an active area of investigation.

2. Radiomics and Texture Analysis:

- Radiomics involves extracting quantitative features from medical images, and texture analysis is a significant component. Researchers are studying how these features can provide valuable information for disease characterization, treatment response assessment, and predicting patient outcomes.

IMAGING ANALYSIS, TECHNIQUES, & AI

3. Multimodal Imaging Integration:

- Integrating information from multiple imaging modalities is a focus of research. Combining data from CT, MRI, PET, and other sources can enhance diagnostic accuracy and provide a more comprehensive understanding of diseases.

4. AI for Early Detection and Screening:

- Researchers are working on AI algorithms for early detection and screening of diseases, particularly in areas like mammography for breast cancer and lung cancer screening using CT scans.

5. Explainable AI (XAI) in Radiology:

- Addressing the interpretability and transparency of AI algorithms is crucial for gaining the trust of radiologists and clinicians. Research is ongoing to develop explainable AI models that provide insights into the decision-making process of AI algorithms.

6. Automated Reporting and Workflow Optimization:

- Automation of reporting tasks and workflow optimization is a growing area of interest. Researchers are developing AI tools to assist radiologists in generating structured reports, prioritizing cases, and improving overall efficiency.

7. Generative Adversarial Networks (GANs):

- GANs are being explored for generating synthetic medical images to augment limited datasets and improve the robustness of AI models. This is particularly important in scenarios where obtaining large labeled datasets is challenging.

IMAGING ANALYSIS, TECHNIQUES, & AI

8. Clinical Decision Support Systems:

- Integration of AI into clinical decision support systems is a key research focus. These systems aim to assist healthcare professionals by providing evidence-based recommendations and improving diagnostic accuracy.

9. Security and Privacy in AI:

- Research is addressing security and privacy concerns associated with medical imaging data. Techniques such as federated learning and homomorphic encryption are being explored to enable collaborative AI research while protecting patient privacy.

10. Quantitative Imaging Biomarkers:

- Identification and validation of quantitative imaging biomarkers for various diseases are ongoing. These biomarkers have the potential to aid in disease diagnosis, prognosis, and treatment planning.

11. Integration with Electronic Health Records (EHR):

- Researchers are exploring ways to integrate AI tools with electronic health records to provide a more comprehensive view of patient data and facilitate seamless healthcare delivery.

12. Global Collaboration and Standardization:

- Efforts are being made to establish international standards for AI in radiology. Collaborative initiatives aim to share datasets, develop benchmarking tools, and create standardized evaluation metrics for AI algorithms.

SEGMENT ANYTHING IN MEDICAL IMAGES

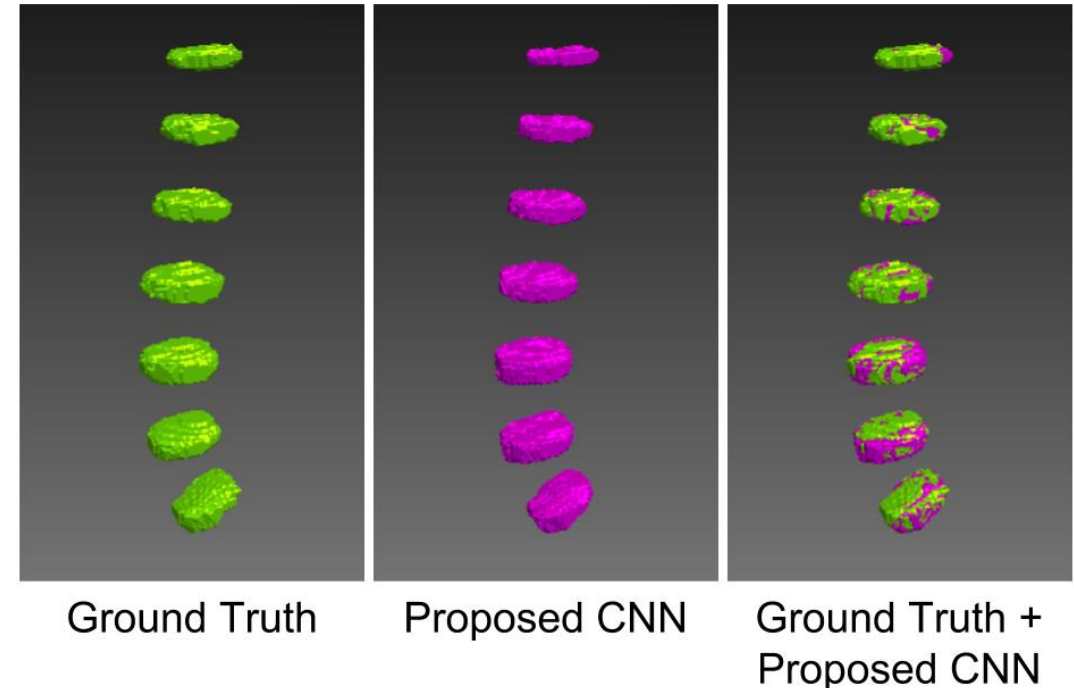
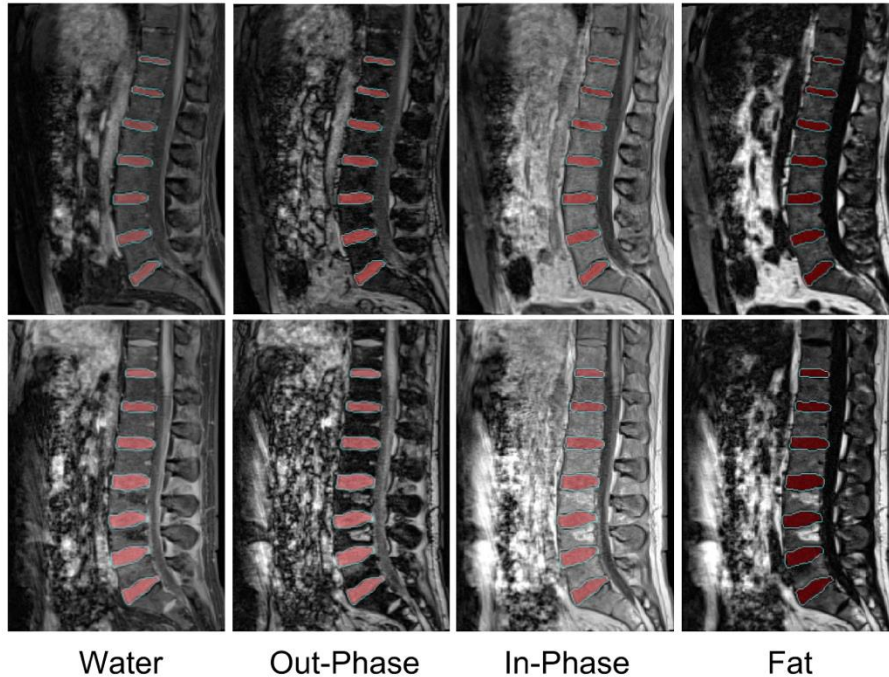
- Medical image segmentation is a critical component in clinical practice, facilitating accurate diagnosis, treatment planning, and disease monitoring. However, existing methods, often tailored to specific modalities or disease types, lack generalizability across the diverse spectrum of medical image segmentation tasks. Here we present MedSAM, a foundation model designed for bridging this gap by enabling universal medical image segmentation. The model is developed on a large-scale medical image dataset with 1,570,263 image-mask pairs, covering 10 imaging modalities and over 30 cancer types. We conduct a comprehensive evaluation on 86 internal validation tasks and 60 external validation tasks, demonstrating better accuracy and robustness than modality-wise specialist models. By delivering accurate and efficient segmentation across a wide spectrum of tasks, MedSAM holds significant potential to expedite the evolution of diagnostic tools and the personalization of treatment plans.
- [Segment anything in medical images | Nature Communications](#)

MEDICAL IMAGE SEGMENTATION

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MEDICAL IMAGE SEGMENTATION

- **Medical Image Segmentation** is a computer vision task that involves dividing an medical image into multiple segments, where each segment represents a different object or structure of interest in the image. The goal of medical image segmentation is to provide a precise and accurate representation of the objects of interest within the image, typically for the purpose of diagnosis, treatment planning, and quantitative analysis.
- [Medical Image Segmentation | Papers With Code](#)





RESEARCH AND DATA ANALYSIS

A POPULATION HEALTH PERSPECTIVE ON ARTIFICIAL INTELLIGENCE

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC7323781/>
- The burgeoning field of Artificial Intelligence (AI) has the potential to profoundly impact the public's health. Yet, to make the most of this opportunity, decision-makers must understand AI concepts. In this article, we describe approaches and fields within AI and illustrate through examples how they can contribute to informed decisions, with a focus on population health applications. We first introduce core concepts needed to understand modern uses of AI and then describe its sub-fields. Finally, we examine four sub-fields of AI most relevant to population health along with examples of available tools and frameworks. Artificial intelligence is a broad and complex field, but the tools that enable the use of AI techniques are becoming more accessible, less expensive, and easier to use than ever before. Applications of AI have the potential to assist clinicians, health system managers, policy-makers, and public health practitioners in making more precise, and potentially more effective, decisions.

A POPULATION HEALTH PERSPECTIVE ON ARTIFICIAL INTELLIGENCE

- Defining intelligence has proven to be controversial and challenging. While the debate continues, most acknowledge that humans are intelligent beings. This means that, to some extent, acting intelligently can be defined as acting in a way that is indistinguishable from how a human would act. This generally accepted criterion has been named the Turing test and passing this test was hypothesized to require the skills listed in Table below.

Skills required to pass the Turing test

Natural language processing (NLP) to enable communication with an interlocutor

Machine learning (ML) to adapt to new circumstances and detect or extrapolate patterns

Knowledge representation to store what is known or sensed

Computer vision to perceive objects and understand images

Computer vision to perceive objects and understand images

Robotics to manipulate objects and move about

THE ROLE OF AI AND PREDICTIVE ANALYTICS IN POPULATION HEALTH MANAGEMENT

- [AI & Predictive Analytics for Population Health Management](#)
- **EHR Data utilization:** Leveraging large datasets from electronic health records to develop insights into patient populations, treatment efficacy, and disease trends. AI models are being refined to extract actionable insights that can enhance clinical decision-making and patient management.
- **Data Analytics Platform**

Data Source	Types of Data
Electronic Health Records	Clinical Data, Lab results, Demographics
Insurance Claims	Health utilization, costs, outcomes
Social Determinants of Health	Income, education, housing, geographic data

EVALUATION OF ARTIFICIAL INTELLIGENCE TECHNIQUES IN DISEASE DIAGNOSIS AND PREDICTION

- [Evaluation of artificial intelligence techniques in disease diagnosis and prediction - PMC](#)
- **Population Health Analytics:** Researching how AI can analyze population health data to identify trends, predict outbreaks, and improve public health responses. The focus is on using diverse datasets to inform healthcare policies and optimize resource allocation.
- A broad range of medical diagnoses is based on analyzing disease images obtained through high-tech digital devices. The application of artificial intelligence (AI) in the assessment of medical images has led to accurate evaluations being performed automatically, which in turn has reduced the workload of physicians, decreased errors and times in diagnosis, and improved performance in the prediction and detection of various diseases. AI techniques based on medical image processing are an essential area of research that uses advanced computer algorithms for prediction, diagnosis, and treatment planning, leading to a remarkable impact on decision-making procedures. Machine Learning (ML) and Deep Learning (DL) as advanced AI techniques are two main subfields applied in the healthcare system to diagnose diseases, discover medication, and identify patient risk factors. The advancement of electronic medical records and big data technologies in recent years has accompanied the success of ML and DL algorithms. ML includes neural networks and fuzzy logic algorithms with various applications in automating forecasting and diagnosis processes. DL algorithm is an ML technique that does not rely on expert feature extraction, unlike classical neural network algorithms. DL algorithms with high-performance calculations give promising results in medical image analysis, such as fusion, segmentation, recording, and classification. Support Vector Machine (SVM) as an ML method and Convolutional Neural Network (CNN) as a DL method is usually the most widely used techniques for analyzing and diagnosing diseases. This review study aims to cover recent AI techniques in diagnosing and predicting numerous diseases such as cancers, heart, lung, skin, genetic, and neural disorders, which perform more precisely compared to specialists without human error. Also, AI's existing challenges and limitations in the medical area are discussed and highlighted.

ARTIFICIAL INTELLIGENCE, MACHINE LEARNING AND GENOMICS

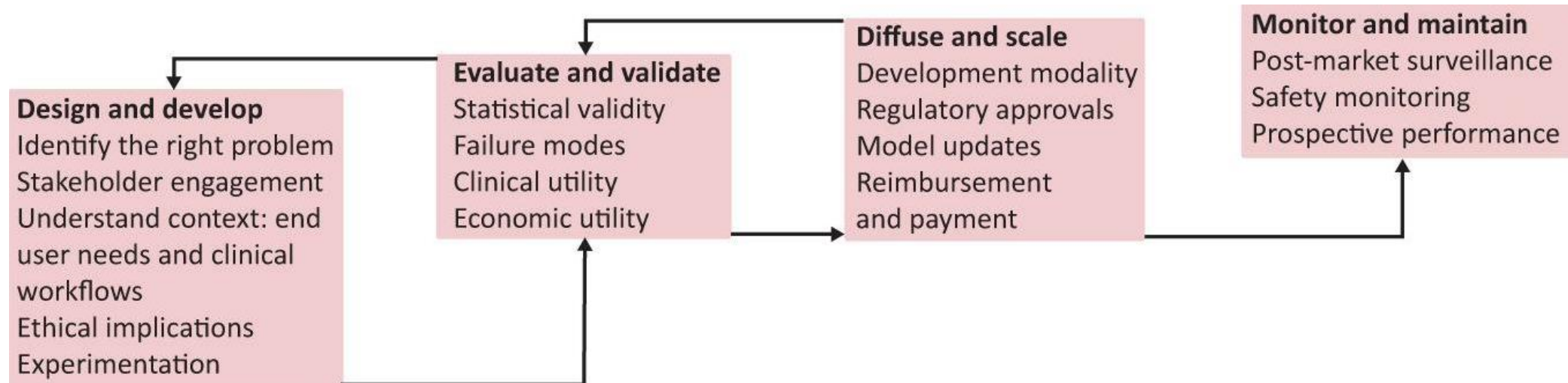
- <https://www.genome.gov/about-genomics/educational-resources/fact-sheets/artificial-intelligence-machine-learning-and-genomics>
- **The Big Picture**
 - The genomics field continues to expand the use of computational methods such as artificial intelligence and machine learning to improve our understanding of hidden patterns in large and complex genomics data sets from basic and clinical research projects.
 - Machine learning analyses could benefit disease research and genomic tools like CRISPR.
 - NHGRI is identifying and shaping its unique role in the convergence of genomic and machine learning research.

ARTIFICIAL INTELLIGENCE IN HEALTHCARE: TRANSFORMING THE PRACTICE OF MEDICINE

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8285156/>
- Integration of Genomic Data
- Artificial intelligence (AI) is a powerful and disruptive area of computer science, with the potential to fundamentally transform the practice of medicine and the delivery of healthcare. In this review article, we outline recent breakthroughs in the application of AI in healthcare, describe a roadmap to building effective, reliable and safe AI systems, and discuss the possible future direction of AI augmented healthcare systems.

ARTIFICIAL INTELLIGENCE IN HEALTHCARE: TRANSFORMING THE PRACTICE OF MEDICINE

- Multi-step, iterative approach to build effective and reliable AI-augmented systems in healthcare



ARTIFICIAL INTELLIGENCE IN HEALTHCARE: TRANSFORMING THE PRACTICE OF MEDICINE

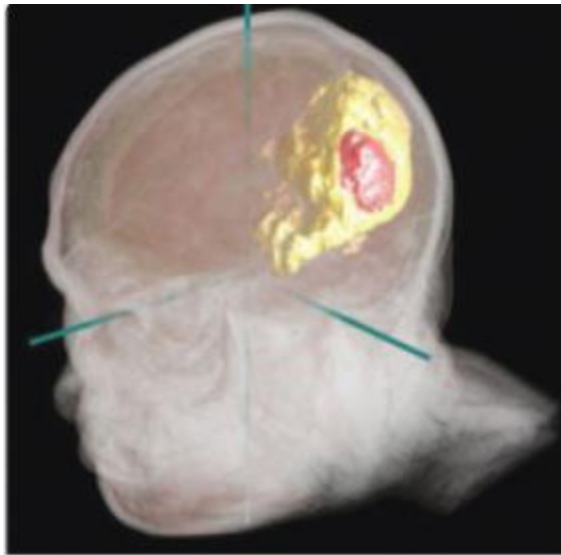
■ Widescale adoption and application of artificial intelligence in healthcare

Timeline	Connected/augmented care	Precision diagnostics	Precision therapeutics	Precision Medicine	Summary
Short term: 0-5 years	Internet of things in healthcare Virtual assistants Augmented telehealth Personalised mental health support	Precision imaging (eg diabetic retinopathy and radiotherapy planning)	CRISPR (increasing use)	Digital and AI enabled research hospitals ³⁰	AI automates time consuming, high-volume repetitive tasks, especially within precision imaging
Medium-term: 5-10 years	Ambient intelligence in healthcare	Large-scale adoption and scale-up of precision imaging	Synthetic biology Immunomics	Customisation of healthcare Robotic assisted therapies	AI uses multi-modal datasets to drive precision therapeutics
Long term: >10 years	Autonomous virtual health assistants, delivering predictive and anticipatory care Networked and connected care organisations (single digital infrastructure)	Holographic and hybrid imaging Holomics (integrated genomic/radiomic/proteomic/clinical/immunohistochemical data)	Genomics medicine AI driven drug discovery	New curative treatments AI empowered healthcare professionals (eg digital twins)	AI enables healthcare systems to achieve a state of precision medicine through AI-augmented healthcare and connected care

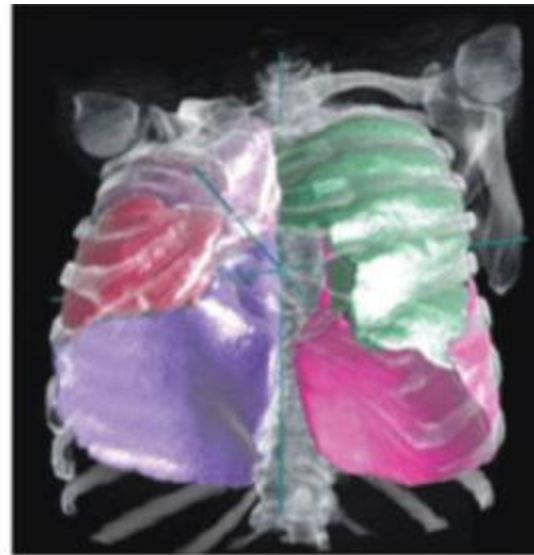
Timings are illustrative to widescale adoption of the proposed innovation taking into account challenges / regulatory environment / use at scale.

ARTIFICIAL INTELLIGENCE IN HEALTHCARE: TRANSFORMING THE PRACTICE OF MEDICINE

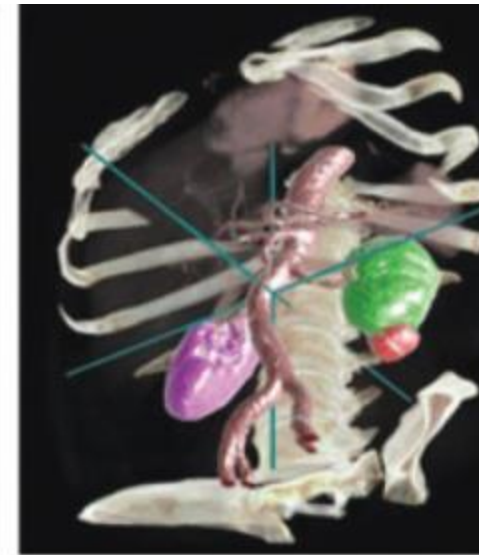
- Potential applications for the InnerEye deep learning toolkit include quantitative radiology for monitoring tumour progression, planning for surgery and radiotherapy planning



Quantitative radiology



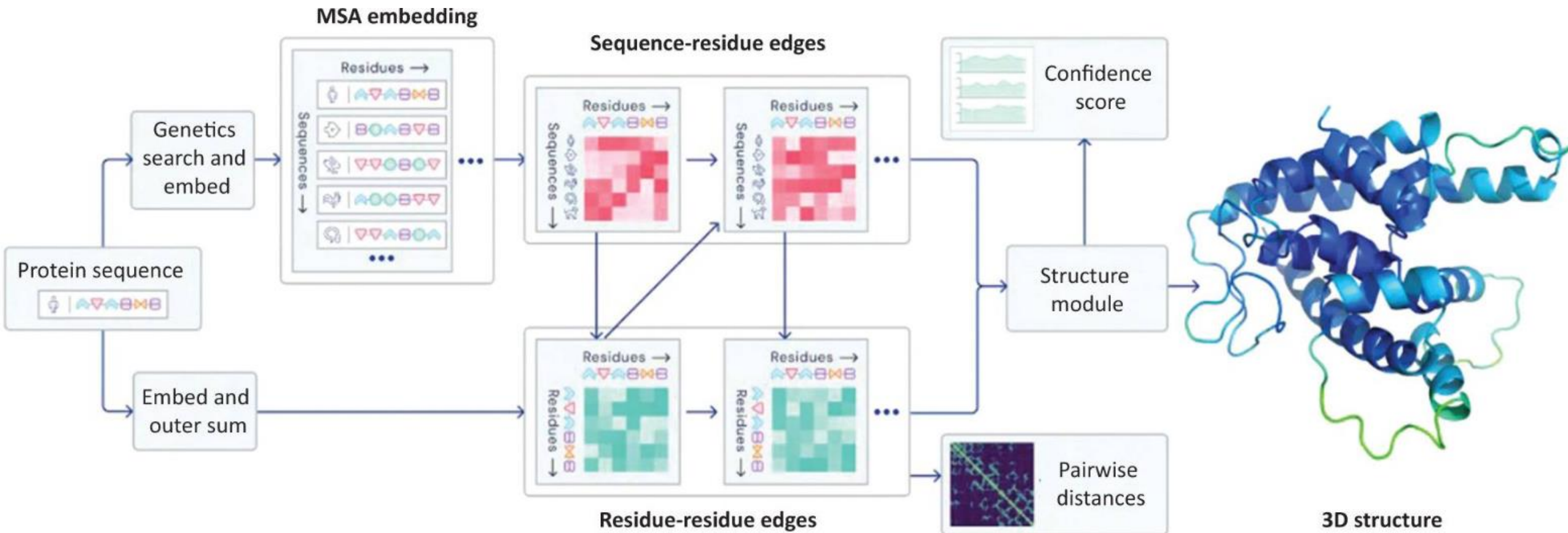
Radiation oncology



Surgical planning

ARTIFICIAL INTELLIGENCE IN HEALTHCARE: TRANSFORMING THE PRACTICE OF MEDICINE

- An overview of the main neural network model architecture for AlphaFold. MSA = multiple sequence alignment.



QUOTES FROM TECHNOLOGY LEADERS

- Satya Nadella, chief executive officer, Microsoft: 'AI is perhaps the most transformational technology of our time, and healthcare is perhaps AI's most pressing application.'
- Tim Cook, chief executive officer, Apple: '[Healthcare] is a business opportunity ... if you look at it, medical health activity is the largest or second-largest component of the economy.'
- Google Health: 'We think that AI is poised to transform medicine, delivering new, assistive technologies that will empower doctors to better serve their patients. Machine learning has dozens of possible application areas, but healthcare stands out as a remarkable opportunity to benefit people.'



THANKS!